

Introduction



What is
computer vision?



What a person sees

The goal of computer vision is
to give computers
(super) human-level perception

Computer Vision

Make computers understand images and video.



What kind of scene?

Where are the cars?

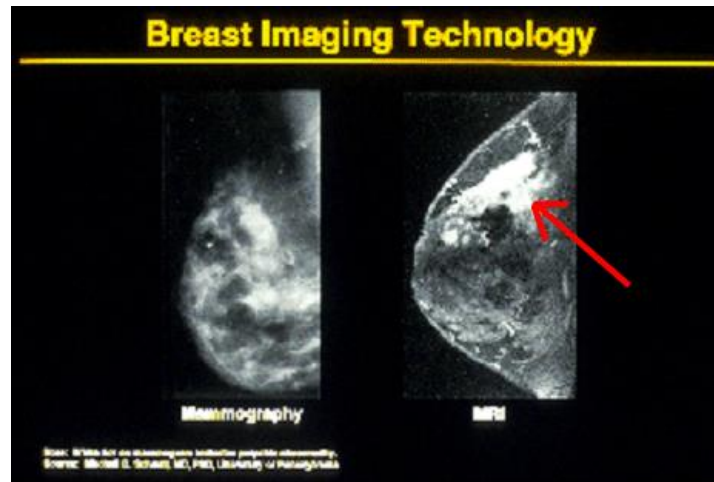
How far is the building?

...

Why computer vision matters



Safety



Health



Security

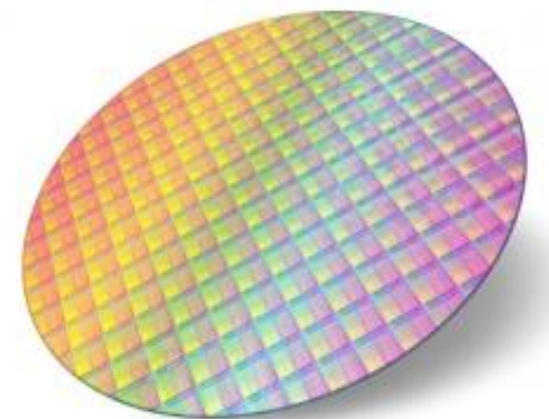


Fun

Applications of computer vision

Machine vision

Automated visual inspection



Object Recognition



Toshiba Tech IS-910T

2013



DataLogic LaneHawk LH4000

2012

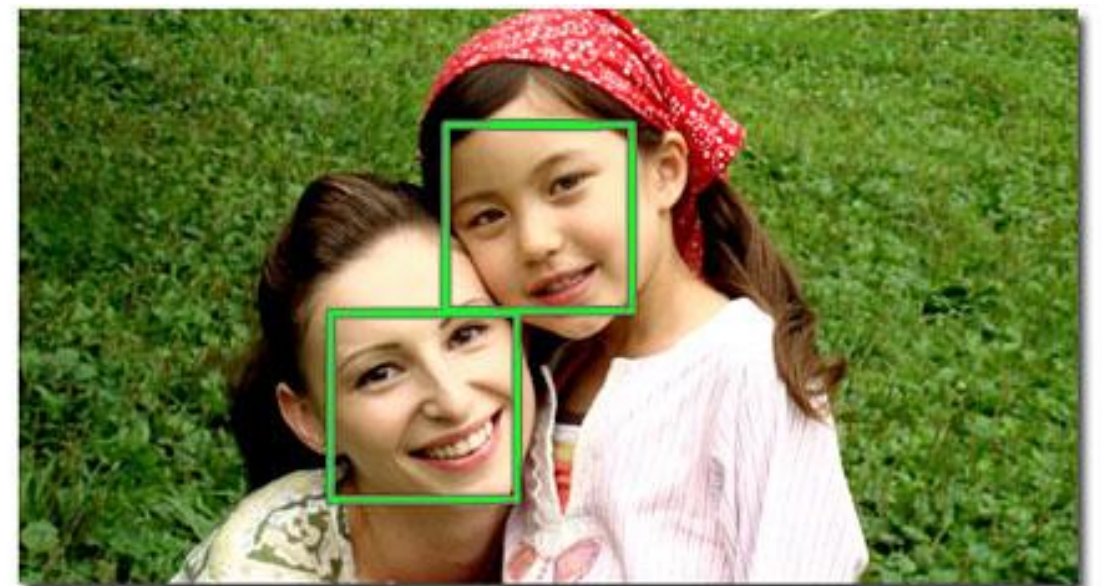
Face detection



Sony Cyber-shot



Age recognition

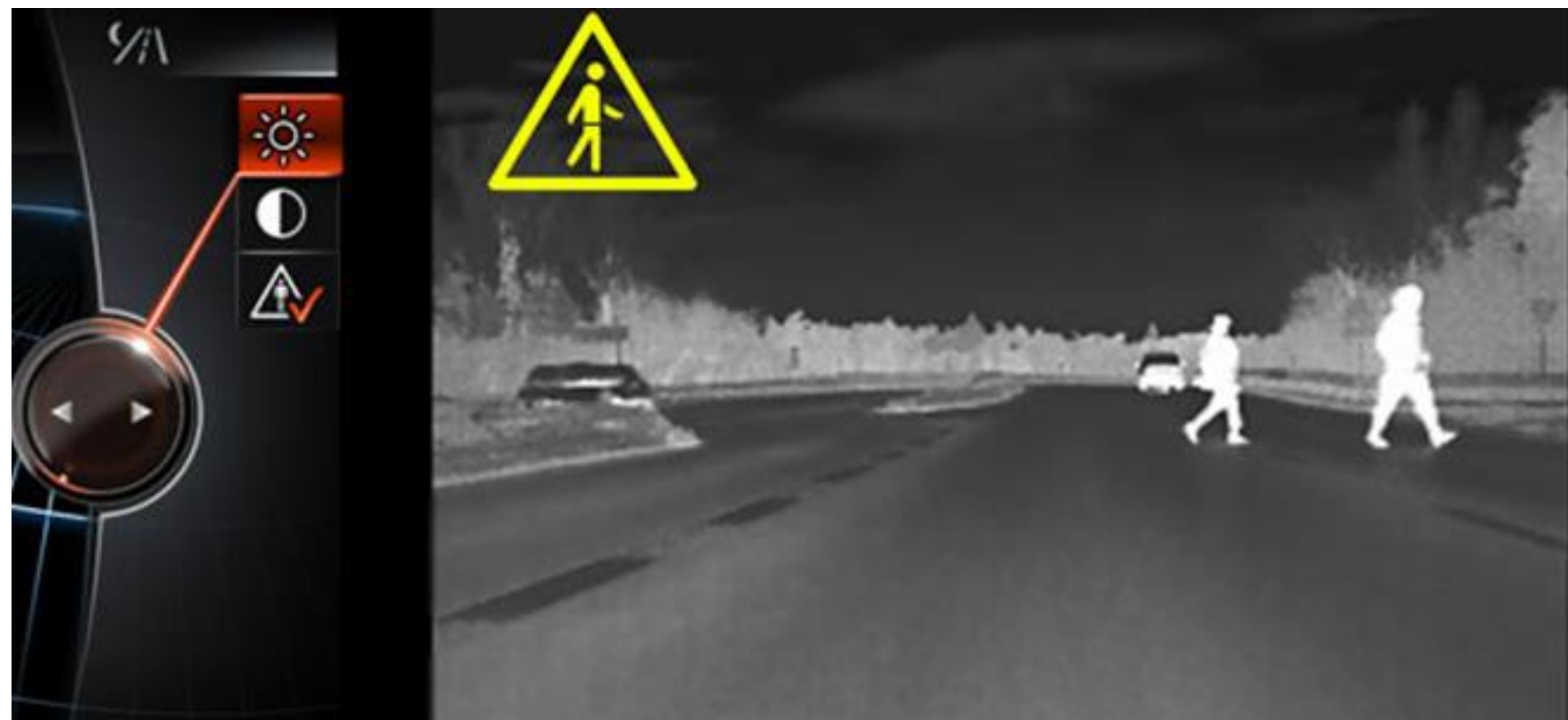


Smile recognition



BMW 5 series

BMW night vision



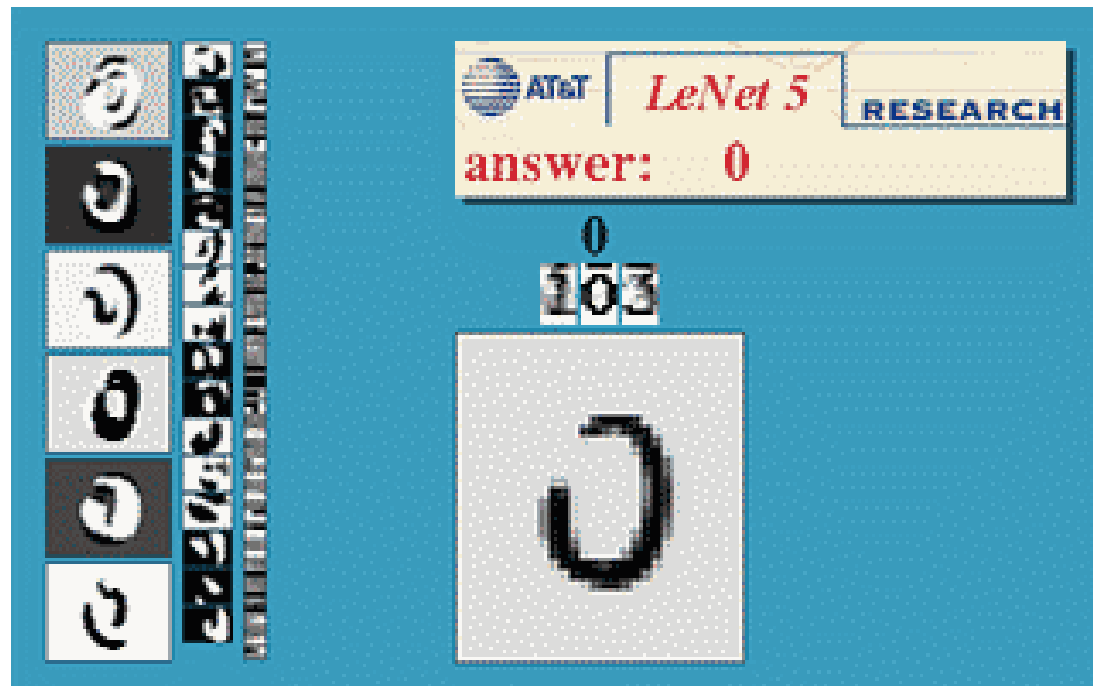
Virtual Fitting



Optical character recognition (OCR)

Technology to convert scanned docs to text

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs

<http://www.research.att.com/~yann/>

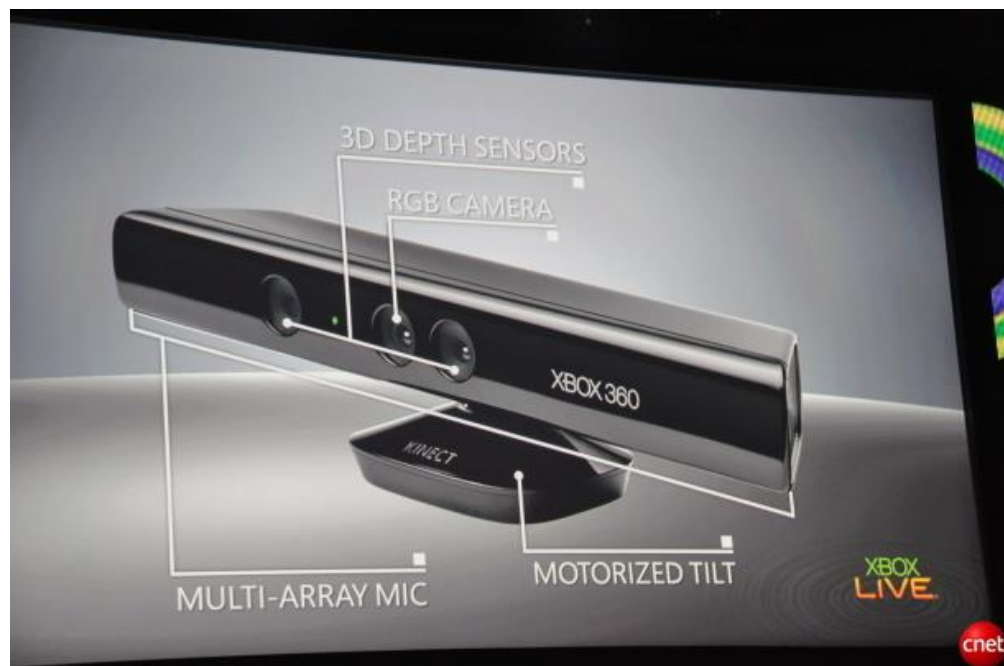


License plate readers

http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

Interactive Games: Kinect

- Object Recognition:
<http://www.youtube.com/watch?feature=iv&v=fQ59dXOo63o>
- Mario: <http://www.youtube.com/watch?v=8CTJL5IUjHg>
- 3D: <http://www.youtube.com/watch?v=7QrnwoO1-8A>
- Robot: <http://www.youtube.com/watch?v=w8BmgtMKFbY>



Contents

Image processing:

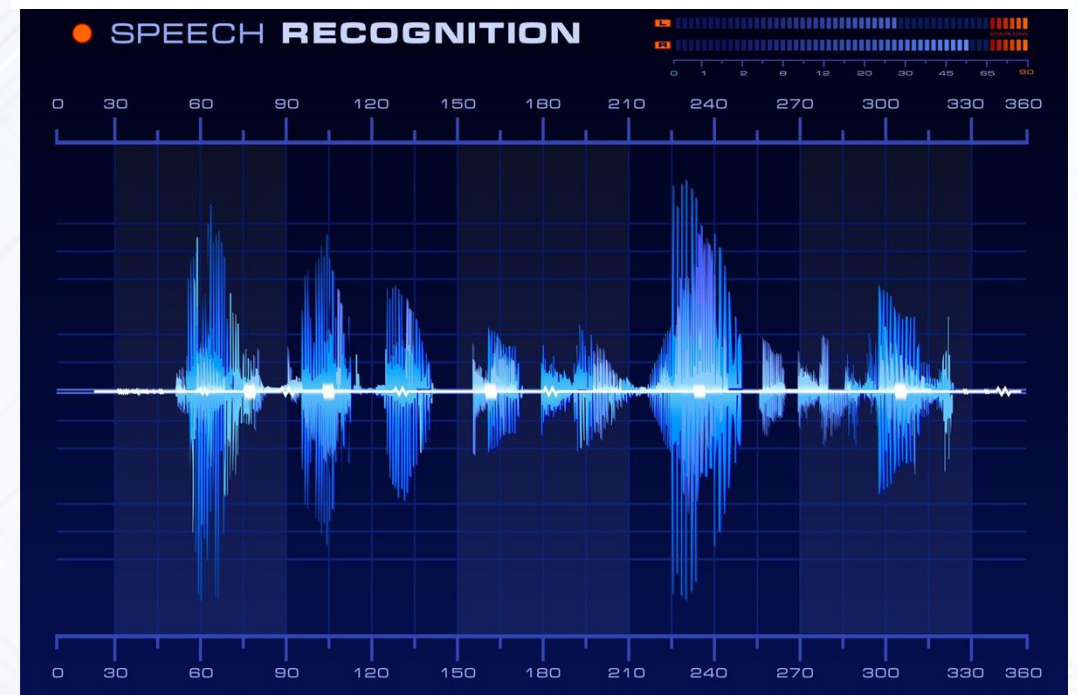
- Basics of filtering.
- Image pyramids.
- Gradients and lines.
- Hough transforms.

INTRODUCTION

What is Deep Learning?

Deep learning is a branch of machine learning that uses data, loads and loads of data, to teach computers how to do things only humans were capable of before.

For example, how do machines solve the problems of perception?

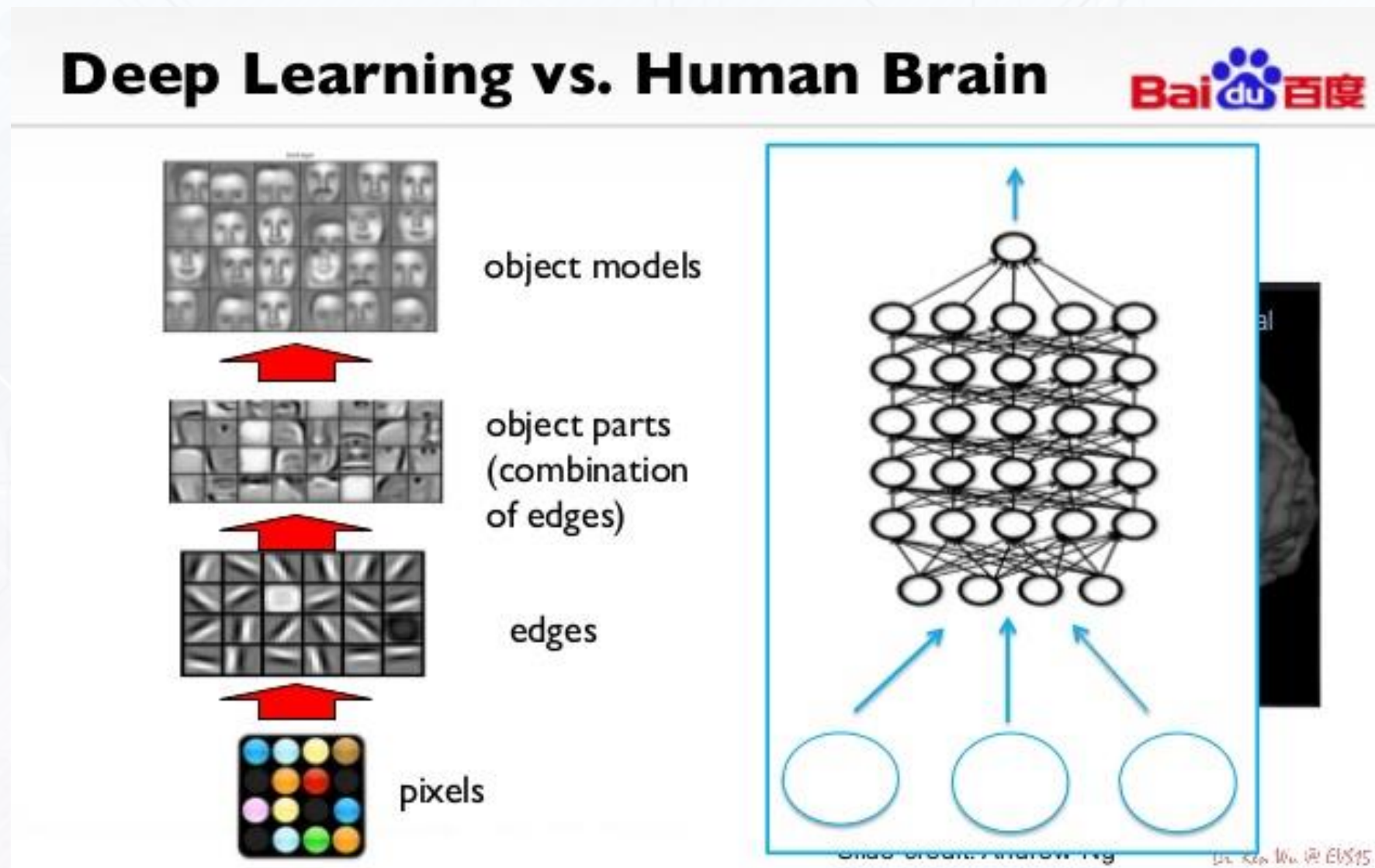


HISTORY

- 1958: Frank Rosenblatt creates the perceptron, an algorithm for pattern recognition.
- 1989: Scientists were able to create algorithms that used deep neural networks.
- 2000's: The term “deep learning” begins to gain popularity after a paper by Geoffrey Hinton.
- 2012: Artificial pattern-recognition algorithms achieve human-level performance on certain tasks.

PRINCIPLE

Deep learning is based on the concept of artificial neural networks, or computational systems that mimic the way the human brain functions.

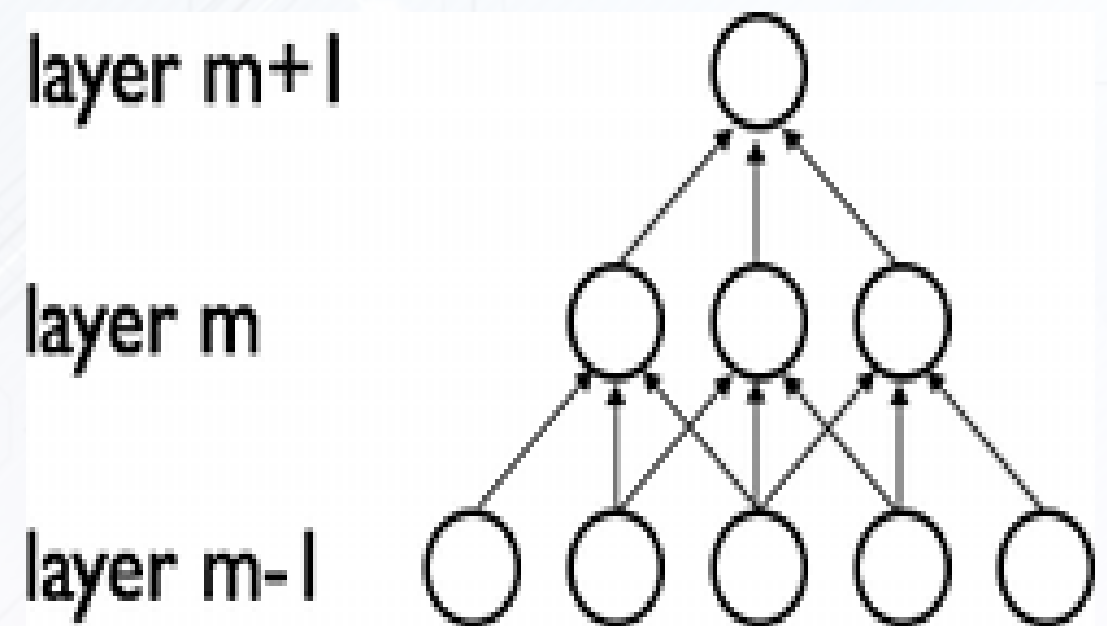


TECHNOLOGY

Deep learning is a fast-growing field, and new architectures, variants appear every few weeks. We'll see discuss the major three:

1. Convolution Neural Network (CNN)

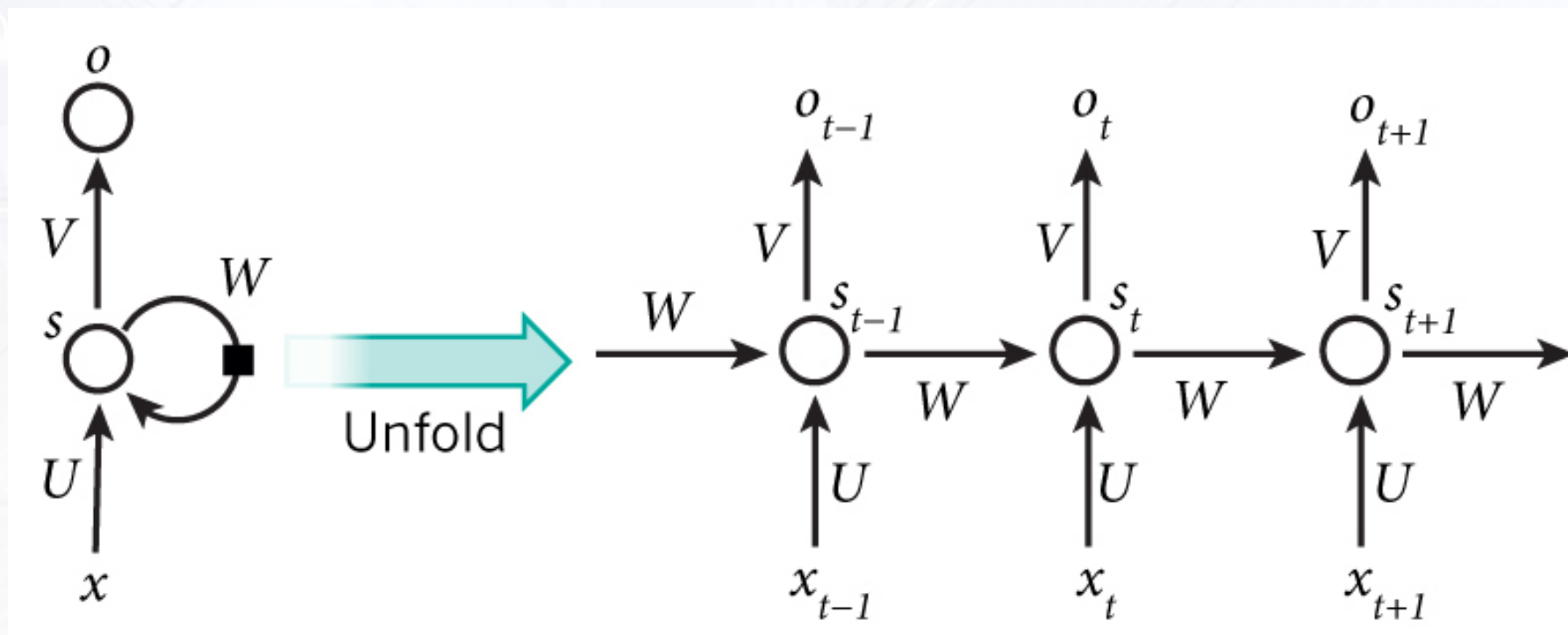
CNNs exploit spatially-local correlation by enforcing a local connectivity pattern between neurons of adjacent layers.



TECHNOLOGY

2. Recurrent Neural Network (RNN)

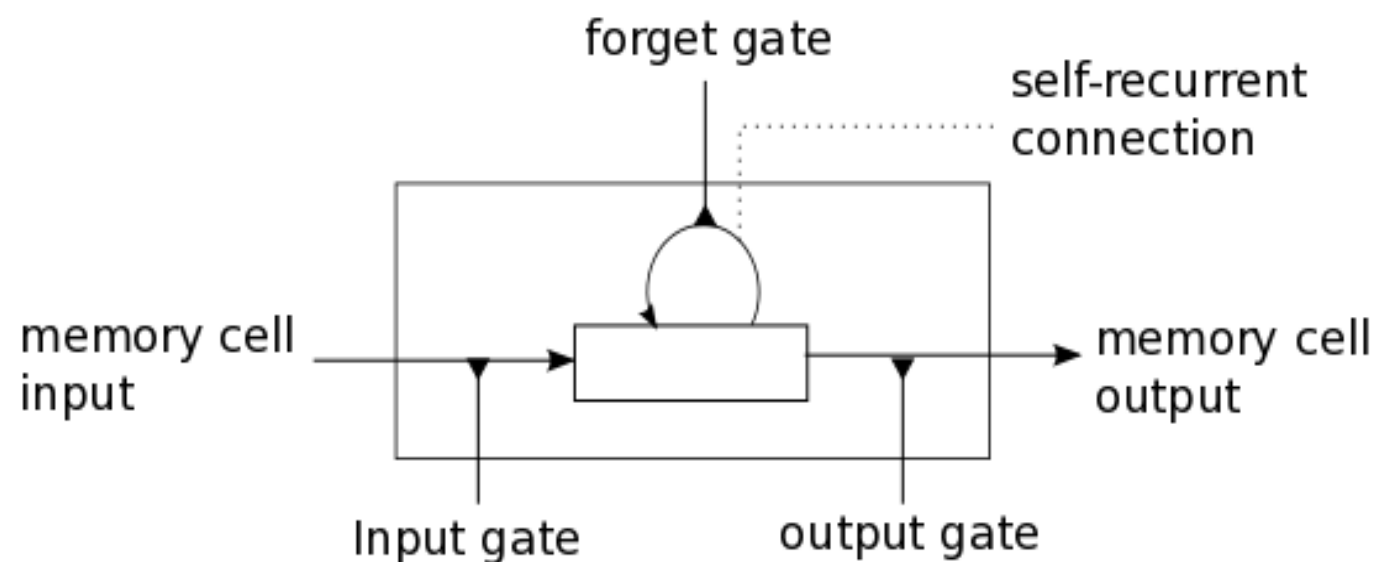
RNNs are called recurrent because they perform the same task for every element of a sequence, with the output being depended on the previous computations. Or RNNs have a “memory” which captures information about what has been calculated so far.



TECHNOLOGY

3. Long-Short Term Memory

LSTM can learn "Very Deep Learning" tasks that require memories of events that happened thousands or even millions of discrete time steps ago. LSTM works even when there are long delays, and it can handle signals that have a mix of low and high frequency components.



WORKING

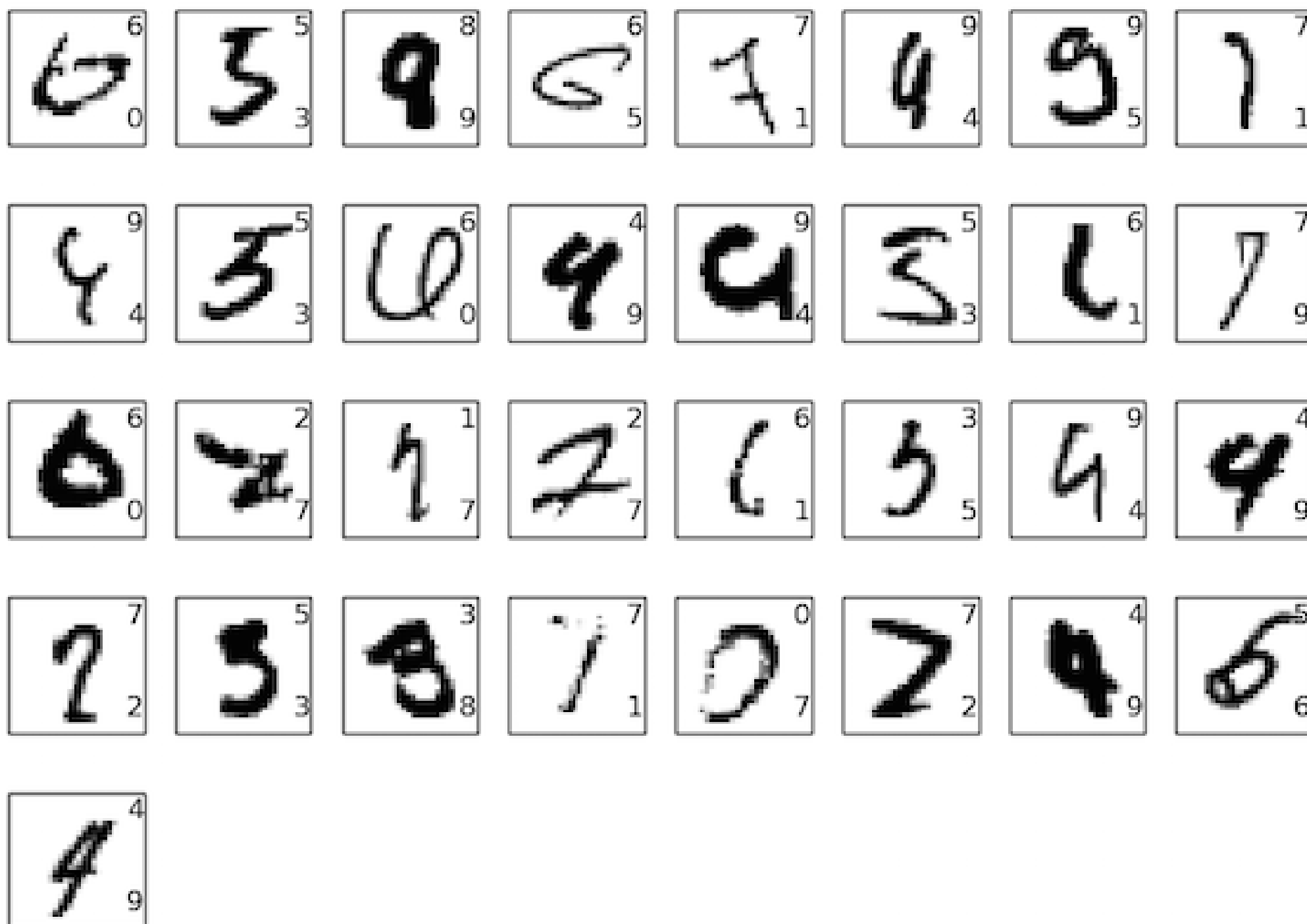
Consider the following handwritten sequence:

A handwritten sequence of digits '504192' displayed on a white rectangular background. The digits are written in a cursive, handwritten style. The '5' is formed with a single stroke, the '0' is a simple oval, the '4' has a vertical stem and a horizontal crossbar, the '1' is a single vertical stroke, the '9' has a loop at the bottom, and the '2' is formed with two strokes.

Most people effortlessly recognize those digits as 504192. That ease is deceptive.

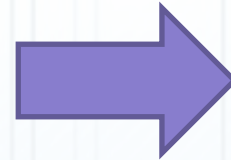
The difficulty of visual pattern recognition becomes apparent if you attempt to write a computer program to recognize digits like those above.

WORKING



WORKING

The idea of neural network is to develop a system which can learn from these large training examples.



0	4	1	9	2	1	3	1	4	3
5	3	6	1	7	2	8	6	9	4
0	9	1	1	2	4	3	2	7	3
8	6	9	0	5	6	0	7	6	1
8	7	9	3	9	8	5	9	3	3
0	7	4	9	8	0	9	4	1	4
4	6	0	4	5	6	1	0	0	1
7	1	6	3	0	2	1	1	7	9
0	2	6	7	8	3	9	0	4	6
7	4	6	8	0	7	8	3	1	5

A training Sample

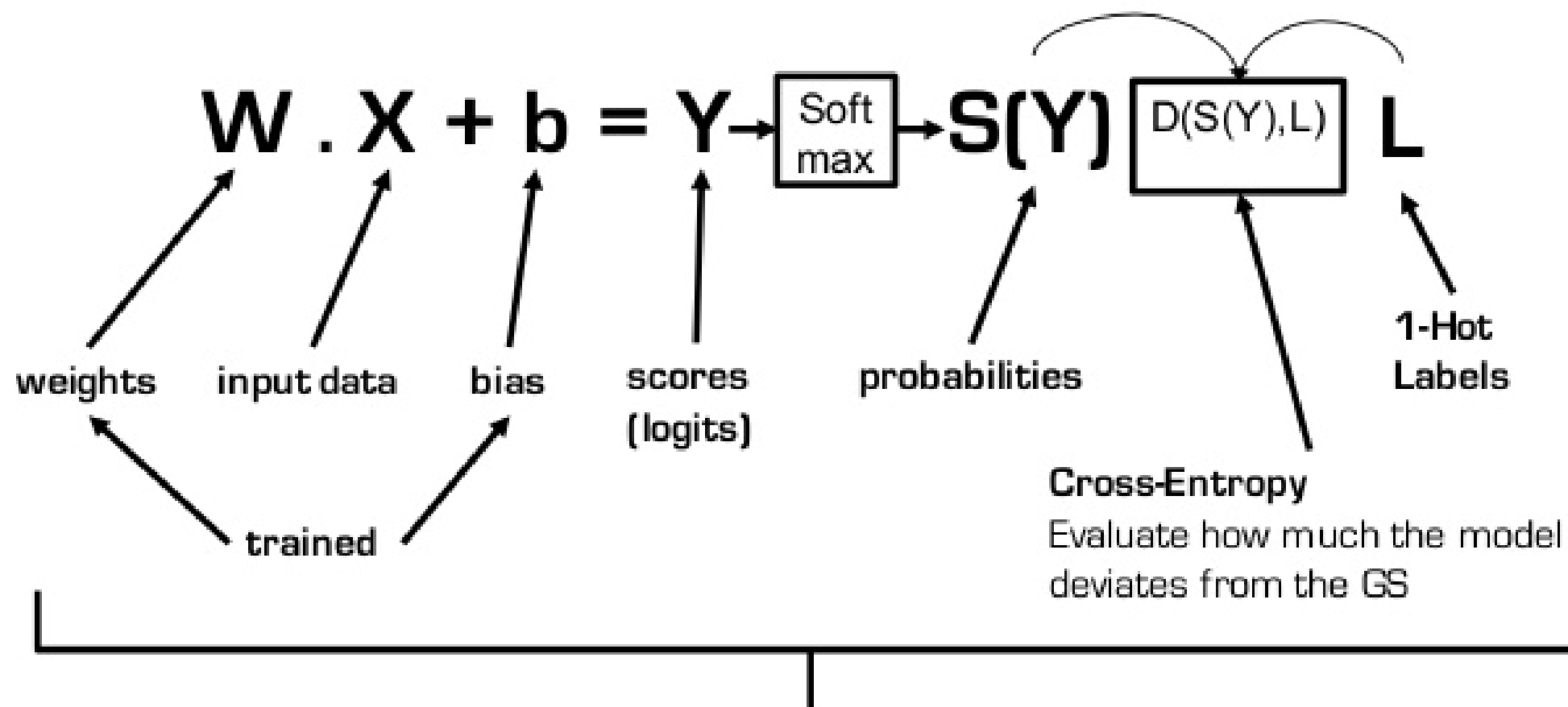
Each neuron assigns a weighting to its input — how correct or incorrect it is relative to the task being performed. The final output is then determined by the total of those weightings

A very basic approach: Binary Classifier

FORMULATIONS

The basis of deep learning is classification which can be further used for detection, ranking, regression, etc.

- **Logistic Classifier => Linear Classifier**



ADVANTAGES

1. It does feature extraction, no need for engineering features
2. Moving towards raw features
3. Better optimization
4. A new level of noise robustness
5. Multi-task and transfer learning
6. Better Architectures

CHALLENGES

1. Need a large dataset
2. Because you need a large dataset, training time is usually significant.
3. The scale of a net's weights is important for performance. When the features are of the same type this is not a problem. However, when the features are heterogeneous, it is.
4. Parameters are hard to interpret—although there is progress being made.
5. Hyperparameter tuning is non-trivial.

REAL TIME APPLICATIONS

1. Automatic Colorization of Black and White Images



2. Automatically Adding Sounds To Silent Movies

3. Automatic Machine Translation



4. Object Classification and Detection in Photographs

5. Automatic Handwriting Generation

Machine learning Mastery
Machine Learning Mastery

REAL TIME APPLICATIONS

- 6. Automatic Text Generation
- 7. Automatic Image Caption Generation



"girl in pink dress is jumping in air."



"black and white dog jumps over bar."



"young girl in pink shirt is swinging on swing."

- 8. Automatic Game Playing

FUTURE SCOPE

1. Deep Learning will speed search for extra terrestrial life.



RobERt, short for Robotic Exoplanet Recognition for Exoplanets that are beyond our solar system.

FUTURE SCOPE

2. For Astronauts, Next Steps on Journey to Space Will Be Virtual



3. Droughts and Deep Learning: Measuring Water Where It's Scarce

CONCLUSION

The low maturity of Deep Learning and its applications such as large deep neural networks achieve the best results on speech recognition, visual object recognition and several language related task field warrants extensive future research. Nevertheless, the possibilities of deep learning in future are infinite ranging from driverless cars, to robots exploring the universe and to what not if the upcoming architectures are creative enough.